Panel Evidence for the Moment Theory of Bond Pricing

Research Design

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1 Introduction and Relation to the Main Paper

The main paper, *A Moment Theory of Bond Pricing*, proposes a Duration-Centered Theory of valuation in which duration acts as the principal state variable, convexity acts as a second-order correction, and higher-order timing moments determine residual pricing effects, extending the classical duration concept of Macaulay [1]. Section 10 of the main paper translates this structure into a statistical claim: when bonds are regressed cross-sectionally on their timing moments, the coefficient on duration should dominate, the addition of convexity should materially improve fit, and successive higher moments should contribute with monotonically diminishing explanatory power, formalized as $R_1^2 < R_2^2 < R_3^2 < \cdots$ with decreasing increments. That section, however, remains a design sketch. It specifies neither the data with which the regression is to be estimated, nor the panel structure that modern empirical fixed-income research demands, nor the inferential apparatus required to distinguish genuine moment premia from the many econometric artifacts that afflict bond panels.

This supplement closes that gap. We develop a complete panel-data research design—an *empirical companion* to Section 10 of the main paper—consisting of four components. First, we formalize the predictions of the theory as five testable hypotheses (Section 2). Second, we construct a public-source data plan built on FINRA TRACE transaction data, Treasury yield curves distributed through FRED, and the Gürkaynak–Sack–Wright zero-coupon curve, so that the entire exercise is replicable without proprietary terminals (Section 3). Third, we specify the estimation architecture: two-way fixed-effects panel regressions with multiway-clustered inference, Fama–MacBeth cross-sectional regressions with Shanken-corrected standard errors, and a Gibbons–Ross–Shanken test of joint moment-pricing restrictions (Sections 4–6). Fourth, we provide a robustness suite and a fully documented Python pipeline that implements the design end-to-end (Sections 7–8).

Two principles govern the design. The first is *honesty about identification*: for a given bond, duration and higher timing moments evolve largely mechanically as the bond ages, so the fixed-effects and cross-sectional estimators answer different questions, and we state precisely which estimator identifies which object. The second is *no fabricated evidence*: because this supplement precedes data collection, all result tables are specification tables that record predicted signs, test statistics to be computed, and pre-committed decision rules, rather than invented estimates. The design is pre-registration-ready in the sense of clinical and empirical finance research practice.

2 Testable Hypotheses

The theory of the main paper implies five hypotheses. Throughout, the timing moments of bond i at date t are denoted $D_{i,t}$ (duration), $C_{i,t} \equiv \mu_{2,i,t}$ (convexity), and $\mu_{k,i,t}$ for $k \geq 3$ (higher central moments), all computed from the discounted cash-flow weighting scheme of the main paper. The first three hypotheses concern *explanatory hierarchy*—the relative contribution of each moment to the pricing regression. The remaining two concern *economic structure*—moment premia in returns and the no-arbitrage clustering argument of Proposition 1 of the main paper.

Hypothesis 1 (Duration Dominance)**.** In cross-sectional regressions of log prices (or, with sign reversal, yields) on the timing-moment vector, the duration coefficient is the dominant source of explanatory power: R^2 from duration alone accounts for the majority of total explained variation attributable to the moment vector.

Hypothesis 2 (Convexity Increment)**.** Conditional on duration, convexity carries incremental pricing content: the added coefficient β_C is statistically significant, economically meaningful, and of the sign implied by the cumulant expansion, $\text{sign}(\beta_C) > 0$ in a log-price specification evaluated at positive yields.

Hypothesis 3 (Monotone Hierarchy with Diminishing Increments)**.** Ordering the regressors as (D), (D, C), (D, C, μ_3), (D, C, μ_3, μ_4), …, the sequence of adjusted R^2 is strictly increasing, and the increments $\Delta_k = R^2_k - R^2_{k-1}$ decline monotonically in k.

Hypothesis 4 (Moment Premia)**.** Expected excess returns on bond portfolios load significantly on moment factors: the duration factor earns a positive premium (compensation for interest-rate risk), the convexity factor earns a premium consistent with its optionality interpretation, and higher-moment factors earn smaller, possibly statistically indistinguishable from zero, premia.

Hypothesis 5 (Duration Clustering)**.** Among duration-matched bonds, conditional price dispersion declines with the similarity of convexity: $|P_A - P_B|$ within a duration bin is an increasing function of $|C_A - C_B|$, consistent with the Duration-Arbitrage Bound (Proposition 1 of the main paper).

Table 1 maps each hypothesis to the econometric machinery of Sections 4–6. Note that the hypotheses are ordered logically: Hypotheses 1–3 concern the pricing surface and are testable with prices alone, Hypothesis 4 requires returns, and Hypothesis 5 requires neither returns nor premia, only cross-sectional dispersion—making it the cheapest test and a natural first diagnostic.

Table 1: Mapping from hypotheses to tests.

Hypothesis	Primary test	Decision rule
H1 (Duration Dominance)	FM cross-section, M1 vs. M5	Share of moment-attributable R^2 from D exceeds 50%
H2 (Convexity Increment)	Two-way FE, M2 vs. M1	t-test on $\hat{\beta}_C > 0$ at 5% level, clustered SEs
H3 (Hierarchy)	Sequential nested models	Bootstrap test of monotone declining Δ_k
H4 (Moment Premia)	GRS on moment-sorted portfolios	Joint $\alpha = 0$ rejected with correct factor signs
H5 (Clustering)	Within-bin dispersion regression	Slope on $\lvert C_A - C_B \rvert$ positive and significant

3 Data and Sample Construction

3.1 Public Data Sources

The design deliberately restricts itself to sources that are free or universally available through academic subscriptions, so that any laboratory can replicate the panel. Corporate and agency transaction prices come from FINRA TRACE (Trade Reporting and Compliance Engine), available to researchers through WRDS; Treasury prices and characteristics come from the CRSP Monthly Treasury database or, in a minimal variant, from the Daily Treasury Par Yield Curve published by the U.S. Department of the Treasury; the risk-free discount curve and its history come from the Gürkaynak–Sack–Wright (GSW) zero-coupon yield dataset [10], distributed through the Federal Reserve Bank of St. Louis FRED system. Macroeconomic controls, when required for the dynamic extension, come from FRED as well. Table 2 summarizes the sources, the variables each provides, and their roles in the design.

Table 2: Public data sources for the panel.

Source	Variables	Role	Access
FINRA TRACE	CUSIP-level transaction prices, volume, trade date	Security-level prices for the dependent variable	WRDS
CRSP Treasury	Treasury prices, coupons, maturities, call features	Government panel; cash-flow schedules	WRDS
U.S. Treasury (treasury.gov)	Daily par yields, auction data	Minimal-cost alternative to CRSP	Free
GSW curve (via FRED)	Zero-coupon yields, 1M–30Y, 1961–present	Discounting weights for moments	Free
FRED	Macro controls (VIX, term spreads, recessions)	Sub-sample conditioning, robustness	Free

3.2 Sample Filters

Raw TRACE panels are noisy: odd lots, reported errors, and cancelled trades generate spurious price variation. The panel applies a standard sequence of filters, each recorded in the pipeline configuration file so the sample is exactly reproducible. We retain fixed-coupon, non-callable, non-putable, bullet bonds; we drop bonds with less than two trades in a calendar month; we winsorize transaction prices at the 1st and 99th percentiles within CUSIP-month; we exclude bonds priced below 20 or above 150 percent of par, where reporting errors concentrate; and we exclude the last month of a bond's life, where illiquidity and accretion effects contaminate the moment–price relation. For the government panel, the analogous filters draw on CRSP flags for callable and flower-bond issues.

3.3 Constructing the Timing Moments

For each bond-month, the discounted cash-flow weights are computed using the contemporaneous GSW curve rather than a flat yield, with the flat-yield variant retained as a robustness exercise. With cash flows $(c_{i,j}, \tau_{i,j})$ at times $\tau_{i,j}$ remaining, and a spot curve $y_t(\tau)$, the price and weights are

$$P_{i,t} \;=\; \sum_{j} c_{i,j}\, e^{-y_t(\tau_{i,j})\,\tau_{i,j}}, \qquad w_{i,j,t} \;=\; \frac{c_{i,j}\, e^{-y_t(\tau_{i,j})\,\tau_{i,j}}}{P_{i,t}}, \tag{1}$$

and the timing variable $T_{i,t}$ takes value $\tau_{i,j}$ with probability $w_{i,j,t}$. Duration, convexity, and higher central moments follow as

$$D_{i,t} = \mathbb{E}_{i,t}[T], \qquad C_{i,t} = \mathrm{Var}_{i,t}(T), \qquad \mu_{k,i,t} = \mathbb{E}_{i,t}\big[(T - D_{i,t})^k\big],\ k \geq 3. \tag{2}$$

The dependent variable for the pricing regressions is the log of the clean price per unit notional, $\log P_{i,t}$; the yield-to-maturity computed from the same price serves as the sign-reversed alternative. A critical measurement decision is the *monthly frequency*: moments drift smoothly within a month, so monthly sampling preserves nearly all cross-sectional information while avoiding double-counting of trades of the same bond.

4 Panel Estimation Design

4.1 Specifications

The core panel regression estimates the pricing surface V of the main paper's bond space $\mathcal{B} = (D, C, \mu_3, \mu_4, \ldots)$ under increasingly demanding assumptions. The pooled baseline is

$$\log P_{i,t} \;=\; \alpha + \beta_D D_{i,t} + \beta_C C_{i,t} + \beta_3 \mu_{3,i,t} + \beta_4 \mu_{4,i,t} + \varepsilon_{i,t}. \tag{3}$$

The fixed-effects specification adds bond effects λ_i and time effects δ_t:

$$\log P_{i,t} \;=\; \beta_D D_{i,t} + \beta_C C_{i,t} + \beta_3 \mu_{3,i,t} + \beta_4 \mu_{4,i,t} + \lambda_i + \delta_t + u_{i,t}, \tag{4}$$

estimated by within transformation. Standard errors are clustered two-way by bond and by calendar month, following Petersen [5] and Cameron, Gelbach, and Miller [6], because both cross-sectional correlation (common yield shocks within a month) and serial correlation (bond-level persistent pricing errors) are first-order features of the panel. Table 3 lists the specification ladder used for the hierarchy test.

Table 3: Specification ladder for the hierarchy test. Predicted signs follow the cumulant expansion $\log P = -yD + \frac{y^2}{2}\kappa_2 - \frac{y^3}{6}\kappa_3 + \cdots$ evaluated at positive yields.

Model	Regressors	Effects	Predicted result
M1	D	none (pooled)	$\hat{\beta}_D < 0$; dominant R^2
M2	D, C	none	$\hat{\beta}_C > 0$; large Δ_2
M3	D, C, μ_3	none	small positive Δ_3
M4	D, C, μ_3, μ_4	none	$\Delta_4 < \Delta_3$
M5	full vector	$\lambda_i + \delta_t$	tests within-bond dynamics of $u_{i,t}$
M6	structural restriction, Eq. (5)	none	fit loss from imposing theory

4.2 The Structural Restriction

The cumulant representation of the main paper, $P = \exp\!\big(-yD + \sum_{n\geq 2}(-y)^n \kappa_n / n!\big)$, implies not merely that moments belong in the pricing regression but that their coefficients are *known functions of the yield*. With $\kappa_2 = \mu_2$, $\kappa_3 = \mu_3$, and $\kappa_4 = \mu_4 - 3\mu_2^2$, truncating after the fourth cumulant gives the restricted specification

$$\log P_{i,t} \;=\; -y_t D_{i,t} + \frac{y_t^2}{2}\, C_{i,t} - \frac{y_t^3}{6}\, \mu_{3,i,t} + \frac{y_t^4}{24}\left(\mu_{4,i,t} - 3C_{i,t}^2\right) + \varepsilon^R_{i,t}, \tag{5}$$

which contains *no free slope parameters*. Model M6 therefore predicts prices out of sample with zero estimated coefficients, and its predictive performance against M4 is a sharp test of the cumulant theory itself, in the spirit of non-nested comparisons [7]. If the restriction holds, M6 should nearly match M4 despite having no estimated slopes; a large fit deficit localizes the failure to the imposed coefficient structure rather than to the choice of moments.

4.3 Identification: What Each Estimator Learns

An honest panel design must confront the fact that, for a fixed bond, the moment vector evolves almost deterministically. As bond i ages, its cash-flow schedule shortens mechanically, so $D_{i,t}$ declines, $C_{i,t}$ declines, and higher moments shift in equally mechanical ways. Within-bond variation of the regressors is therefore largely a deterministic aging profile, and the fixed-effects estimator in Eq. (4) identifies β from deviations of moments from their bond-specific aging paths—variation that is informative about the local curvature of the pricing function along the aging trajectory, but that cannot separately identify a *cross-sectional* pricing surface. Time effects δ_t absorb the common yield movements that would otherwise masquerade as moment premia.

The Fama–MacBeth estimator of Section 5 uses the complementary, between-bond variation: at each date, bonds with different cash-flow profiles occupy different points of the moment space, and the cross-sectional regression traces the pricing surface itself. The two estimators are thus complements rather than competitors, and the design pre-commits to reporting both: the fixed-effects specification answers *how pricing errors evolve within bonds*, while the Fama–MacBeth specification answers *how prices vary across the moment manifold at a point in time*. This division of labor mirrors the distinction between time-series and cross-sectional identification in term-structure regressions [9].

5 Fama–MacBeth Cross-Sectional Regressions

The Fama–MacBeth procedure [2] replaces the single pooled regression with one regression per date. For each month t in the sample, estimate the cross-sectional regression

$$\log P_{i,t} \;=\; \alpha_t + \hat{\beta}_{D,t} D_{i,t} + \hat{\beta}_{C,t} C_{i,t} + \hat{\beta}_{3,t} \mu_{3,i,t} + \hat{\beta}_{4,t} \mu_{4,i,t} + e_{i,t}, \qquad i = 1, \dots, N_t, \tag{6}$$

where N_t is the number of bonds with valid data in month t. The Fama–MacBeth estimate and its uncertainty are then

$$\hat{\beta} \;=\; \frac{1}{T} \sum_{t=1}^{T} \hat{\beta}_t, \qquad \mathrm{SE}(\hat{\beta}) \;=\; \sqrt{\widehat{\mathrm{LRV}}(\hat{\beta}_t - \bar{\beta})/T}, \tag{7}$$

where $\widehat{\mathrm{LRV}}$ denotes the Newey–West long-run variance estimator with lag truncation set to $\lfloor T^{1/3} \rfloor$. Because the regressors are portfolio-invariant realized moments rather than estimated betas, the Shanken correction [4] is not strictly required; it is nevertheless reported as a robustness check because measurement error in the moments generated by the GSW interpolation induces an errors-in-variables problem of similar flavor.

Three design choices deserve emphasis. First, each cross-section is weighted by inverse effective bid-ask spread where available, so that illiquid bonds do not drive the surface; an unweighted variant is retained. Second, the same cross-section is re-estimated with the yield-to-maturity as the dependent variable, for which all predicted signs flip; agreement between the two in absolute value is a specification check on the transformation. Third, the number of laddered maturities within each cross-section is monitored: because government bonds cluster near auction cycles, the effective rank of the moment matrix can deteriorate in particular months, and those months are flagged rather than silently included.

6 Testing the Moment Hierarchy and Moment Premia

6.1 Formal Tests of the Hierarchy

Hypothesis 3 makes two distinct claims: strict improvement ($R^2_k > R^2_{k-1}$) and diminishing increments ($\Delta_k < \Delta_{k-1}$). Because the models are nested, the significance of each increment Δ_k is exactly the Wald test of the coefficients added in moving from model $k-1$ to model k; the design uses the two-way-clustered covariance matrix for this test. The *diminishing* claim is a joint restriction across increments for which no standard closed form exists, so it is evaluated by a pairs bootstrap over bonds: resample bonds with replacement, recompute the full specification ladder, and report the bootstrap distribution of the increment sequence. The pre-committed decision rule is that the theory is supported if the median bootstrap increment is positive at every step and the median sequence is strictly decreasing; partial support is recorded when monotonicity holds but some increment is statistically indistinguishable from zero.

6.2 A GRS Test of Joint Moment Pricing

To test Hypothesis 4, bonds are sorted into N portfolios (deciles by duration, with a cross-sort by convexity within duration deciles), and time-series regressions of portfolio excess returns on the moment factors F_D, F_C, F_3 deliver an intercept vector $\hat{\alpha}$. If the moment factors span the pricing-relevant space, all pricing errors should be jointly zero. The Gibbons–Ross–Shanken statistic [3] is

$$\text{GRS} = \frac{T}{N} \cdot \frac{T-N-M}{T-M-1} \cdot \frac{\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \bar{f}^\top \hat{\Sigma}_f^{-1} \bar{f}}, \tag{8}$$

where M is the number of factors, T the number of months, $\hat{\Sigma}$ the residual covariance matrix, and the denominator the usual finite-sample correction. Rejection of the joint null with correctly signed factor premia supports the factor interpretation of the main paper's Table 5 (duration as the time-preference factor, convexity as the optionality factor); rejection with *incorrect* signs would be evidence against the theory rather than for it, and the design records this distinction explicitly. The linear-regression implementation follows the term-structure pricing literature [9, 8, 11], which provides the natural benchmark for what linear factor regressions on the yield curve can and cannot identify.

6.3 Decision Rules Summary

Table 4 consolidates the inferential decisions. Each row names a statistic, its null, and the pre-committed conclusion under rejection and non-rejection. Pre-commitment matters here: because the moment hierarchy is the central claim of the main paper, the design fixes the decision rules before estimation to prevent specification search from silently reshaping the theory.

Table 4: Pre-committed decision rules.

Statistic	If rejected / condition met	If not rejected / condition fails
Wald on $\hat{\beta}_C$ (M2 vs. M1)	H2 supported	Convexity has no incremental content; theory reduces to duration-only
Bootstrap monotone Δ_k	H3 supported	Hierarchy exists but increments irregular; report pattern
GRS on moment factors	H4 supported if signs correct	Factors priced but misaligned; revisit factor construction
M6 vs. M4 out-of-sample	Cumulant restriction supported if ratio near one	Restriction rejected; estimate free coefficients and diagnose

7 Robustness Architecture

Every headline result is subjected to a fixed battery of robustness exercises, chosen so that each targets a distinct threat to interpretation. Sub-sample stability splits the panel at December 2008, separating the pre-crisis and quantitative-easing regimes in which the discounting environment differs sharply; the theory predicts stability of the *ordering* of moment contributions even if coefficients shift. Rating and maturity-band splits examine whether the hierarchy is an artifact of a particular segment. Placebo moments—the timing moments of randomly permuted cash-flow schedules within each cross-section—should exhibit no pricing content; significance under placebo would indicate an unmodeled mechanical linkage between the regression design and the panel's correlation structure. Winsorization thresholds, the flat-yield versus GSW discounting choice, and alternative clustering dimensions complete the battery. Table 5 lists the dimensions and the failure criterion for each.

Table 5: Robustness dimensions and failure criteria.

Dimension	Variation	Failure criterion
Regime split	Pre/post December 2008	Ordering of moment contributions reverses
Segment split	Rating classes; maturity bands	Hierarchy holds only in one narrow band
Discounting	GSW curve vs. flat yield	Hierarchy depends on curve choice
Outliers	Winsorize at 1/99 vs. 5/95	Hierarchy driven by extreme moments
Placebo	Permuted cash-flow schedules	Placebo moments price significantly
Clustering	One-way vs. two-way clusters	Inference flips with clustering choice

A final robustness dimension concerns the composition of the panel itself. Because TRACE coverage expands dramatically over the sample, the early panel is thin and dominated by large issues; the full-sample results are therefore re-estimated on the balanced sub-panel of bonds present in every month of a rolling window, and on the government-only panel where cash-flow schedules are known with certainty. Convergence of conclusions across these panels is itself evidence, since the sources of measurement error differ substantially between corporate and government data.

8 Implementation: A Public-Source Data Pipeline

The entire design is implemented in a companion Python module, `moment_panel_pipeline.py`, distributed with this supplement. The module is organized as a directed pipeline with five stages, shown in Figure 1: ingestion of raw TRACE, CRSP, and FRED extracts; construction of the cash-flow schedules and timing moments; estimation of the panel and Fama–MacBeth specifications; the hierarchy and GRS test batteries; and the robustness suite. Each stage reads and writes versioned Parquet files, and a single configuration dictionary records every filter threshold, so that the sample is exactly reproducible from raw extracts.



Figure 1: The estimation pipeline. Each stage is a module of the companion Python file and communicates through versioned Parquet files.

Algorithm 1 summarizes the estimation flow. Reproducibility is enforced by three conventions: random seeds are fixed and logged for every bootstrap and placebo exercise; package versions are pinned in an environment file; and each figure and table in the eventual results paper is generated

by a dedicated script that consumes only the versioned intermediate files, never raw data directly. The module includes a self-test mode that runs the full pipeline on a small synthetic panel with known moment structure, allowing the implementer to verify installation correctness before attaching real data.

Algorithm 1: The panel estimation pipeline.

Input: TRACE/CRSP extracts, GSW curve $y_t(\tau)$, configuration $\mathcal{C}$ (filters, thresholds, seeds)
Output: Specification ladder results, GRS statistic, robustness battery

```
Apply filters in C to raw transactions; aggregate to CUSIP-month panel;
foreach bond-month (i, t) do
    Compute discounted weights w_{i,j,t} from Eq. (1) using y_t(τ);
    Compute (D_{i,t}, C_{i,t}, μ_{3,i,t}, μ_{4,i,t}) from Eq. (2);
end
Estimate M1–M4 by pooled OLS; M5 by two-way within transformation;
Evaluate structural model M6 with coefficients fixed by Eq. (5);
foreach month t do
    Estimate cross-section Eq. (6); collect β̂_t
end
;
Aggregate Fama–MacBeth estimates with Newey–West lags; apply Shanken check;
Form moment-sorted portfolios; estimate factor time-series; compute GRS statistic Eq. (8);
Run pairs bootstrap over bonds for monotone-increment test; run placebo and regime splits;
return results tables with pre-committed decision flags;
```

9 Interpretation, Limitations, and Conclusion

9.1 What Would Confirm and What Would Refute

The strongest confirmation of the Duration-Centered Theory would be a triple coincidence: duration dominance in the cross-section (H1), a significant and correctly signed convexity increment with rapidly diminishing higher-order contributions (H2–H3), and joint moment pricing with correct signs in the GRS battery (H4). Such a pattern would elevate the cumulant representation from an algebraic identity to an empirically operative description of the pricing surface, and would justify the moment-feature approach to machine-learning valuation proposed in Section 15 of the main paper. The most informative refutation would be the discovery that convexity dominates duration once the panel is properly conditioned, or that the increments of the hierarchy are irregular in a way that survives the bootstrap—either finding would force a revision of the claim that duration is the unique tangent coordinate of the bond manifold.

9.2 Limitations

Four limitations bound what the design can establish. First, the moment vector is a deterministic transform of the cash-flow schedule and the discount curve, so the regressors are generated regressors; measurement error in the GSW interpolation propagates into all coefficients, and the errors-in-variables direction of the resulting bias is conservative for dominance tests but not for magnitude claims. Second, moments are mutually collinear in any realistic panel—duration and convexity are strongly negatively related across bonds—so individual coefficient magnitudes should be interpreted through the specification ladder's *incremental* R^2 statistics rather than in isolation, and the Shapley decomposition reported in the robustness suite provides the appropriate attribution. Third, the design tests pricing, not causality: a finding that moments price does not establish that investors consciously trade moments of economic time, only that the moment representation organizes the cross-section. Fourth, corporate cash-flow schedules are contractual but not certain, so the corporate panel tests the theory under the assumption

that default risk acts on levels rather than on timing; the credit extension flagged in the main paper's research agenda would relax exactly this assumption.

9.3 Conclusion

This supplement has converted the statistical claim of Section 10 of the main paper into a complete, pre-registration-ready research design: five hypotheses, a public-source data plan, a specification ladder spanning pooled, fixed-effects, structural-restriction, and Fama–MacBeth estimators, a GRS battery for moment premia, and a robustness architecture with pre-committed decision rules. The companion pipeline implements every step in code whose inputs are free data and whose outputs are the exact tables of the decision framework. The design is deliberately austere—no proprietary data, no discretionary filters, no post-hoc rules—because the moment theory's claim to unify fixed-income valuation deserves the same evidentiary discipline as any empirical asset-pricing hypothesis. Its execution is the next step of the research program.

A Data Dictionary

Table 6 defines every variable in the panel. Units are stated explicitly; all price variables are per 100 of notional, and all timing variables are in years.

Table 6: Panel data dictionary.

Variable	Definition	Source	Unit
`cusip_id`	Security identifier, 8-digit CUSIP	TRACE / CRSP	n/a
`date_month`	Calendar month of observation	derived	YYYY-MM
`P_clean`	Median clean transaction price in month	TRACE	per 100 notional
`ytm`	Yield to maturity implied by `P_clean`	derived	annual, decimal
`D`	Present-value weighted mean timing, Eq. (2)	derived	years
`C`	Variance of timing distribution	derived	years^2
`mu3`	Third central moment of timing	derived	years^3
`mu4`	Fourth central moment of timing	derived	years^4
`coupon`	Annual coupon rate	CRSP / Treasury	decimal
`maturity`	Remaining time to final payment	derived	years
`amount_out`	Amount outstanding at month end	CRSP	USD millions
`n_trades`	Trades in month; filter variable	TRACE	count
`ba_spread`	Effective bid-ask estimate	TRACE	per 100 notional
`y_gsw`	GSW zero curve at $\tau = D$	FRED	annual, decimal
`rating`	Composite rating (Moody's/S&P)	WRDS	categorical

B Estimation Code Skeleton

The core estimation functions of the companion module are excerpted below; the full file, including ingestion adapters and the self-test mode, accompanies this supplement as `moment_panel_pipeline.py`.

```
import numpy as np
import pandas as pd

def timing_moments(cashflows, spot_curve):
    """Central moments of discounted timing for one bond-month.

    cashflows : list of (tau_j, c_j) -- times in years, cash amounts
    spot_curve: callable tau -> continuously compounded zero rate
    """
    tau = np.array([t for t, c in cashflows])
    cf = np.array([c for t, c in cashflows])
    pv = cf * np.exp(-spot_curve(tau) * tau)
    w = pv / pv.sum() # discounted weights
    D = float(w @ tau) # duration
    X = tau - D # centered timing
    C = float(w @ X**2) # convexity
    mu3 = float(w @ X**3)
    mu4 = float(w @ X**4)
    return {"D": D, "C": C, "mu3": mu3, "mu4": mu4}

def structural_logprice(mom, y):
    """Cumulant-restricted prediction: no free slope parameters."""
    D, C, m3, m4 = mom["D"], mom["C"], mom["mu3"], mom["mu4"]
    return (-y * D + 0.5 * y**2 * C - y**3 / 6 * m3
            + y**4 / 24 * (m4 - 3 * C**2))

def fama_macbeth(panel, specs, lags=None):
    """Monthly cross-sections, Newey-West averaged betas."""
    rows = []
    for t, g in panel.groupby("date_month"):
        X = g[specs].to_numpy(); X = np.c_[np.ones(len(g)), X]
        yv = g["logP"].to_numpy()
        b = np.linalg.lstsq(X, yv, rcond=None)[0]
        rows.append(pd.Series(b, index=["const"] + specs, name=t))
    B = pd.DataFrame(rows)
    T = len(B)
    if lags is None:
        lags = int(np.floor(T ** (1 / 3)))
    se = B.apply(lambda s: nw_se(s - s.mean(), lags), axis=0) / np.sqrt(T)
    return B.mean(), se

def nw_se(x, lags):
    """Newey-West long-run standard deviation of a demeaned series."""
    g = lags
    L = np.dot(x, x)
    for l in range(1, g + 1):
        w = 1 - l / (g + 1)
        L += 2 * w * np.dot(x[l:], x[:-l])
    return np.sqrt(L / len(x))
```

References

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